The Hong Kong Polytechnic University

School of Professional Education and Executive Development

SEHS4701 Advanced Information Systems Development

2024-25, Semester One

Group Project Assignment

Due on 23rd November 2024

Objectives

The objectives of this project are to:

- Apply J2EE concepts and different programming skills.
- Develop basic web application for an organization by using advanced framework (e.g. Spring boot).
- Solve problems in designed scenario through hands-on activities.

Mission

In this project, a group of about 5 to 6 students is formed as a project team who is going to provide a professional consultancy service for a client on web application design.

Background

Your client is one of Hong Kong Homeless Care (HKHC) non-profit charity organization and it plans to collect donation from public through web channel. Your team is required to prepare a web application regarding to the following requirements:

- Create a web application by using J2EE.
- Its main functions consist of
    1) Collection of a donor (personal or organization) data
    2) A donor can choose which type of donation program and amount of money he wants to support, e.g. $1,000 for homeless children program.
    3) HKHC can view each donation by choosing different selection criteria, e.g. amount larger than $5,000.
    4) A donor can receive an appreciation acknowledgment from a donatee.
- All data will be stored into MySQL database.

Apart from the above, you can make other reasonable functions or assumptions that may affect your design. (e.g. donor registration procedure)

Instruction

Based on the above requirements, your team needs to **write a functional specification** on your design. In the specification, it covers:

- The background and requirements of your client.
- Business Rules
- UI Design
- Database Design
- System Diagram and Workflow
- Test Cases

Submission

Your team is required to submit the following items in softcopy only via the submission link on Blackboard before the deadline. Late submission is not accepted and not marked.

Functional Specification (.pdf) – by Group Leader

The specification summarizes the tasks mentioned above and the corresponding web application design that your team proposed. **Limit the specification to 20 pages.**

Source code and database script (.zip) – by Group Leader

All source code and database schema/data script files are compressed into a single zip file.

Peer-to-peer Evaluation (.docx) – by Everyone

Each student needs to fill in a peer-to-peer evaluation form (to be downloaded from Blackboard) on self-evaluation and evaluating the performance of other members in the same group. The final mark may be adjusted according to individual contribution to the group.

Grading

Aspects	Weight
Problem Analysis (school requirements, assumptions, etc.)	20%
Design and Implementation of Web Application (contents, system flow, program skill, etc.)	60%
Specification Writing (clarity, preciseness, use of English, etc.)	20%
Total	100%

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